Experiment:7 Experimental Characterization of Microwave Passive Components: Directional Coupler, Power Divider, Attenuator and Resonator

Aim: To simulate and characterize microwave passive components using MATLAB by analyzing their S-parameters, attenuation, power division, and resonant frequency.

Apparatus:

Computer/Laptop
MATLAB software

Procedure:

- Open MATLAB and create a new script.
- Define the frequency range and S-parameter values for each component.
- Calculate the transmission and reflection characteristics.
- Simulate the directional coupler, power divider, attenuator and resonator.
- Plot three figures in a single window to compare their characteristics.
- Run the program and observe the results.

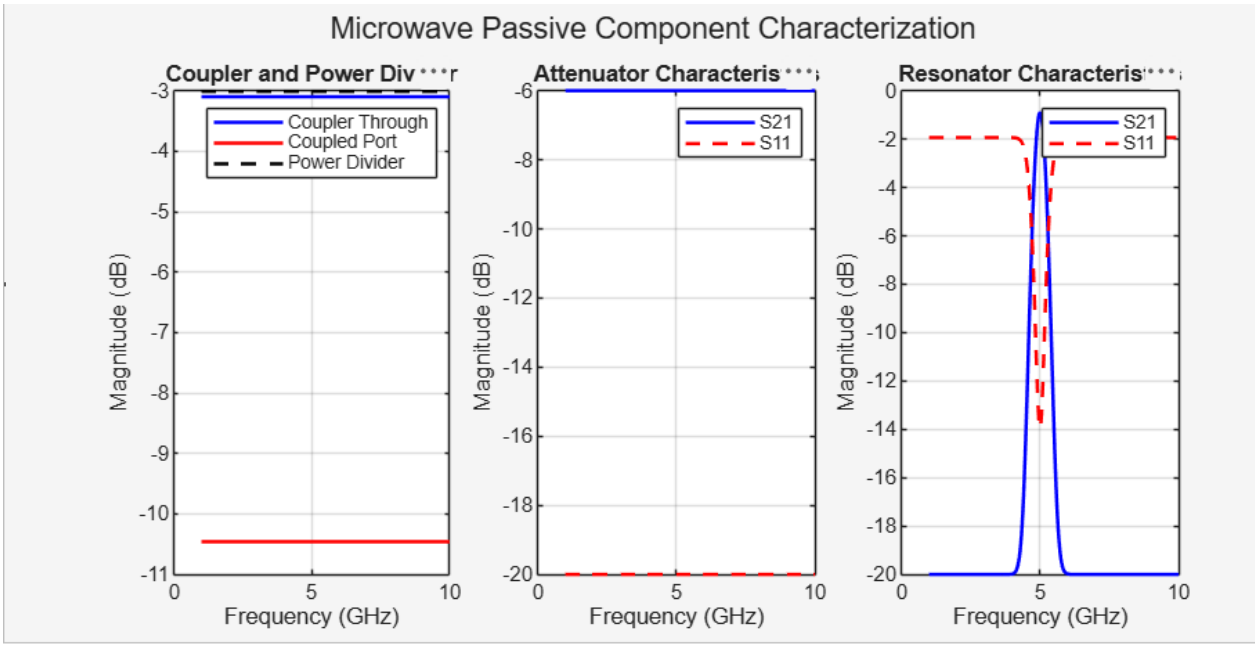
Code:

```
clc;
clear;
close all;
f = linspace(1,10,500);
S21c = 0.7*ones(size(f));
S31c = 0.3*ones(size(f));
S21d = 1/sqrt(2)*ones(size(f));
A = 6;
S21a = 10^(-A/20)*ones(size(f));
```

```
S11a = 0.1*ones(size(f));
f0 = 5;
S21r = 0.1 + 0.8*exp(-((f-f0)/0.35).^2);
S11r = 0.8 - 0.6*exp(-((f-f0)/0.35).^2);
figure;
subplot(1,3,1);
plot(f,20*log10(S21c),'b','LineWidth',1.5);
hold on;
plot(f,20*log10(S31c),'r','LineWidth',1.5);
plot(f,20*log10(S21d),'k--','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Coupler and Power Divider');
legend('Coupler Through','Coupled Port','Power Divider');
subplot(1,3,2);
plot(f,20*log10(S21a),'b','LineWidth',1.5);
hold on;
plot(f,20*log10(S11a),'r--','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Attenuator Characteristics');
legend('S21','S11');
subplot(1,3,3);
plot(f,20*log10(S21r),'b','LineWidth',1.5);
hold on;
```

```
plot(f,20*log10(S11r),'r--','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Resonator Characteristics');
legend('S21','S11');
sgtitle('Microwave Passive Component Characterization');
fprintf('Attenuation = %.1f dB\n',A);
fprintf('Power Divider Output = %.2f dB\n',20*log10(1/sqrt(2)));
fprintf('Resonant Frequency = %.2f GHz\n',f0);
```

Output:



Result:

The microwave passive components were simulated using MATLAB. Their transmission and reflection characteristics were plotted, the power divider showed approximately −3 dB transmission, the attenuator provided 6 dB attenuation, and the resonator showed a response near 5 GHz.